Song Success Prediction Model

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Introduction

The goal of the Song Success Prediction Model was to determine whether a song's pre-release attributes could be used to accurately predict its popularity, aiding artists and labels to make data driven decisions for marketing and release strategies. Two data sources were used as a baseline for this project - Spotify Music Analytics Dataset (2015–2025), and supplemental Country data through the Global Data API. Popularity was defined by the Spotify Analytics dataset, ranging from 0-100.

Throughout this project I follow the full data science workflow: Data Wrangling, Exploratory Data Analysis (EDA), Pre-Processing, and Modeling. Multiple modeling approaches and pivots were explored. Ultimately, the conclusion of this project is that the available features do not provide sufficient predictive signal to build a model that meaningfully outperforms baselines of simple guess framework. This report details that process and the evidence supporting that determination.

Data Wrangling

In the Data Wrangling step, the Spotify Analytics Dataset and Country data were imported and combined in preparation for the Song Success Prediction Model. The data was imported separately into individual dataframes, then cleaned individually prior to merging into one master dataframe.

Spotify Data was imported and contained 85,000 records with 19 attributes. The **popularity** attribute was identified as potentially the target feature for this project as the aim is to create a predictive model that will return a songs' popularity based on the song's attributes. **Stream count** was identified as a feature of interest as it may be an alternative target or it may be necessary to drop prior to model training to prevent leakage.

The other attributes of the Spotify Data represented two types of data related to the song: **Descriptive Data** such as Danceability, Genre, Explicit etc and **Technical Data** such as Duration, Key, Tempo etc. In inspecting the features more carefully, a there were a couple of necessary changes that were made:

- The **release date** column was converted to a datetime object.
- The **duration_ms** column showed the duration of songs in milliseconds, while this data will be ideal for model training, an additional duration column was created titled **duration_mm:ss:ms** to show the duration in a more reader friendly format. This column will be used for the EDA and data visualization only, then will be dropped prior to model training to avoid training on duplicate data.

After the datatypes were corrected and new fields added, the dataset was inspected for **duplicate values**. Inspection revealed that the data frame showed a full 85,000 unique values for track id, indicating that there were no duplicate records. This was further investigated through track name which revealed that the only near duplicate values were re-releases of the same song by artists with different musical attributes. This justified that these records should remain in the dataset.

Then the dataset was inspected for **Missing Values**. The dataset contained 21 missing values for **track_name** and 46 missing values for **album_name**. These were handled as follows:

- **Track Name -** Filled with the sting ‘Unknown’. Determination was made to preserve as many features as possible, and early stages of song production is likely to mean that unnamed songs will be passed though the practical use of the final model. Prior to filling these values, the dataset was searched for any songs already titled “Unknown” with 0 records.
- **Album Name -** Dropped. These 46 records were dropped because unlike track name, there is not a tied feature (track_id) that indicates these albums are separate. If filled with ‘unknown’ as track name was, models may create fake relationships between these tracks and skew results.

After missing values were filled and dropped, and duplicate values were inspected, the spotify dataframe had 84,954 records remaining.

Country Data was imported through the apicountries.com API. To minimize the api call, a list was created from the spotify data of unique country values. This list was used to pull data for only the countries represented from the API. To ensure a clean and accurate query, some of the country names had to be edited to exactly match the name in the api. Once imported the country data included 37 attributes for the 10 countries queried. To clean and standardize the Country data, string columns were normalized to lowercase to ensure consistency, and numeric columns were converted to numeric data types.

The **gini** attribute contained one missing value for India. This column pertaining to a country's economic distribution will likely be quite important to our model, so it was important to fill this value accurately. Filling with a mean or median from the other values would be extremely misleading and inaccurate. So, to fill this value I researched the gini for India and manually filled the missing value with 25.5, a number obtained directly from the World Bank data that is the main source of information for gini. It can be reviewed here: https://data.worldbank.org/indicator/SI.POV.GINI.

Deeper inspection of the Country Data revealed that several of the columns values were comprised of lists and lists of dictionaries. Not only would these attributes not be usable for training the model, they were also difficult to read and would not provide much to any insight in the EDA phase. They each had to be delt with individually:

- The **timezones** and **borders** columns both contained lists of all the timezones/bordering countries for the specific country record. It was determined that the specific

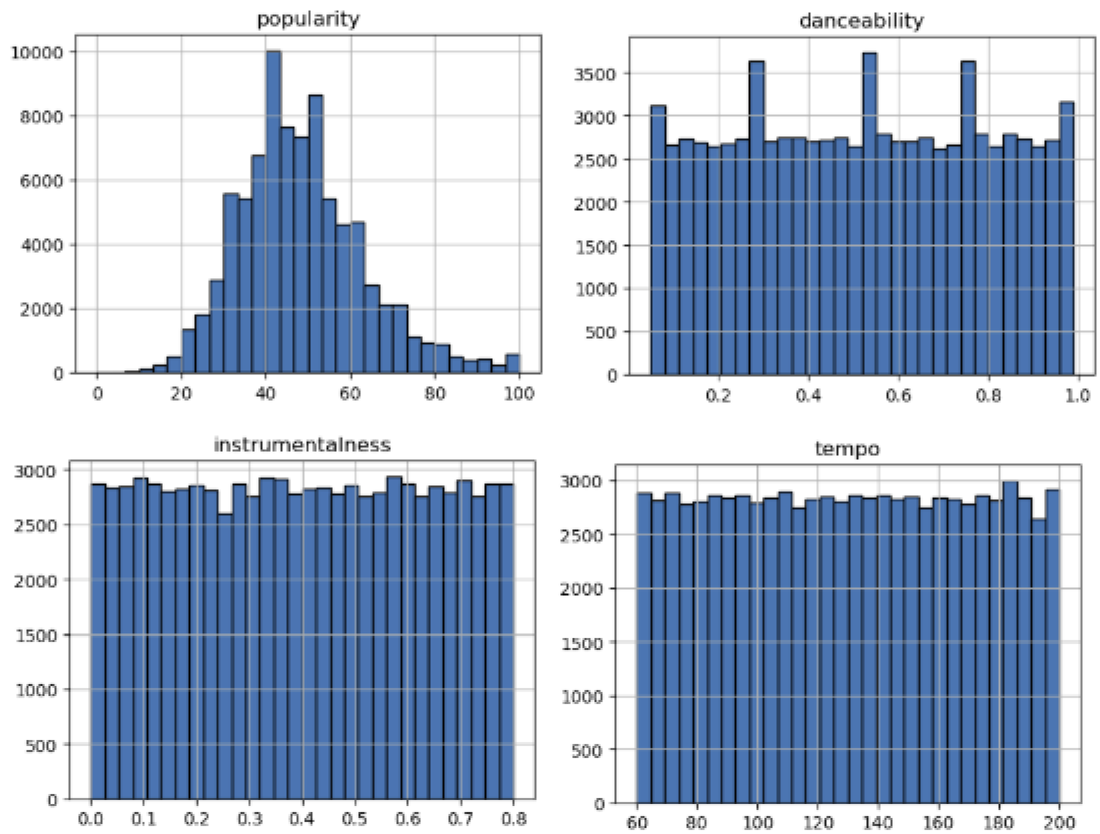
timezones/bordering countries were not needed, however the total count of these attributes could be useful. So these columns were replaced with **num_timezones** and **num_borders** containing the total count for each.
- The **currencies** column contained a list of a dictionary that showed the currency code, currency name, and currency symbol for the country. It was determined that only one of these was needed and the currency code would be the most useful. So this column was replaced with a **currency_code** column.
- The **languages** and **regionalBlocs** columns both contained lists of dictionaries that contained all official languages spoken in the country and all unions/associations the country was affiliated with. **Languages** was replaced with a column showing the total number of languages spoken, as well as unique columns for each language with boolean values showing if the language is an official language of the country. **RegionalBlocs** was replaced with unique columns for each association with boolean values indicating if the country is a part of the association.

After cleaning, several columns were identified as redundant or irrelevant and were dropped from the Countries dataframe. These columns were: topLevelDomain, callingCodes, altSpellings, flag, flags.svg, flags.png, all translation columns, alpha3Code (duplicate information for alpha2Code), timezones (replaced by num_timezones), borders (replaced by num_boarders), nativeName (duplicate information for name), numericCode (duplicate information for name), currencies (replaced by currency_code), languages (replaced by num_languages and individual lang columns), regionalBlocs (replaced by individual bloc columns), language_names (created as a step to create individual lang columns), regional_blocs (created as a step to create individual bloc columns).

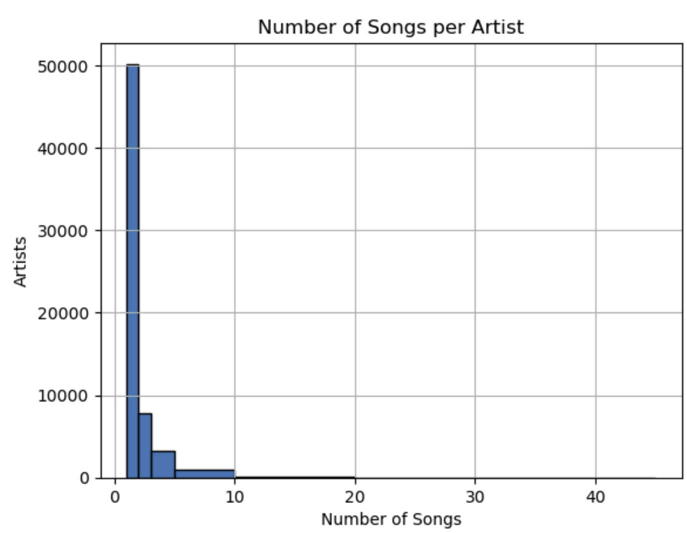
The Spotify Data and the Country Data was then merged into one master dataframe titled **combined_df** fully prepared for Exploratory Data Analysis.

Exploratory Data Analysis

In the Exploratory Data Analysis step of this project, I explored many of the features in the dataset and was able to identify several noteworthy insights from the data. To start, the data was separated into two variables, numerical features and categorial features, then each of those was visualized to gain an understanding of the distribution. The target feature, **popularity**, showed a uniform distribution with a small spike at 100. Surprisingly, features like danceability, instrumentalness, and tempo showed a uniform distribution, which indicated that the dataset has a good variety of songs to train the model. Stream count was heavily right skewed and needed to be log-scaled for a better visualization where it showed that the majority of songs had less than 2.5 million streams with a maximum number of recorded streams at 20 million.



After exploring summary visualizations and distributions, I explored specific features more closely.



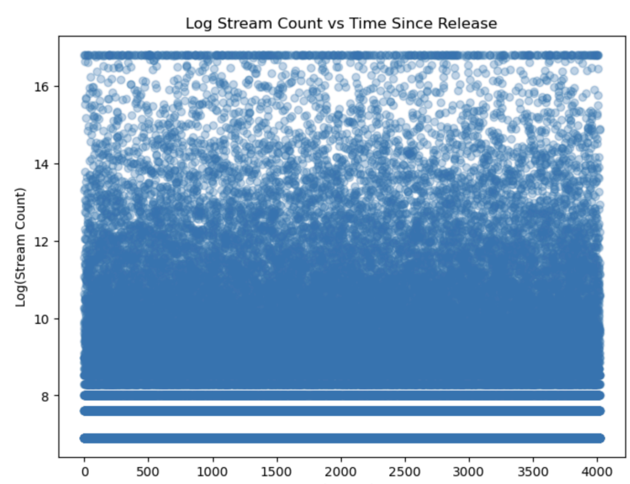
Artists - Artists warranted specific exploration as an assumption can be made that artist reputation and popularity would have a high correlation to song popularity. Surprisingly, the vast majority of artists represented in the dataset only had one song, and therefore only one record, meaning there were almost 30 artists with a perfect 100 mean and median popularity score and the maximum result of 20 million for stream count. This indicated that artist popularity could not be accurately calculated since the majority of artists would automatically equal their one song's popularity score resulting in leakage of the popularity feature.

	artist_name	num_songs	num_albums	mean_popularity	median_popularity	total_streams
46066	Nicholas Graham	1	1	100.0	100.0	14458000
58190	Thomas Park	1	1	100.0	100.0	20000000
22635	Harold Price	1	1	100.0	100.0	20000000
38645	Luis Waters	1	1	100.0	100.0	1708000
28522	Jodi Bishop	1	1	100.0	100.0	20000000
19472	Elizabeth Stanley	1	1	100.0	100.0	8383000
25095	Janet Ross	1	1	100.0	100.0	4952000
6646	Bobby Gomez	1	1	100.0	100.0	20000000
21985	Glenn Murphy	1	1	100.0	100.0	20000000
55217	Stanley Baldwin	1	1	100.0	100.0	20000000

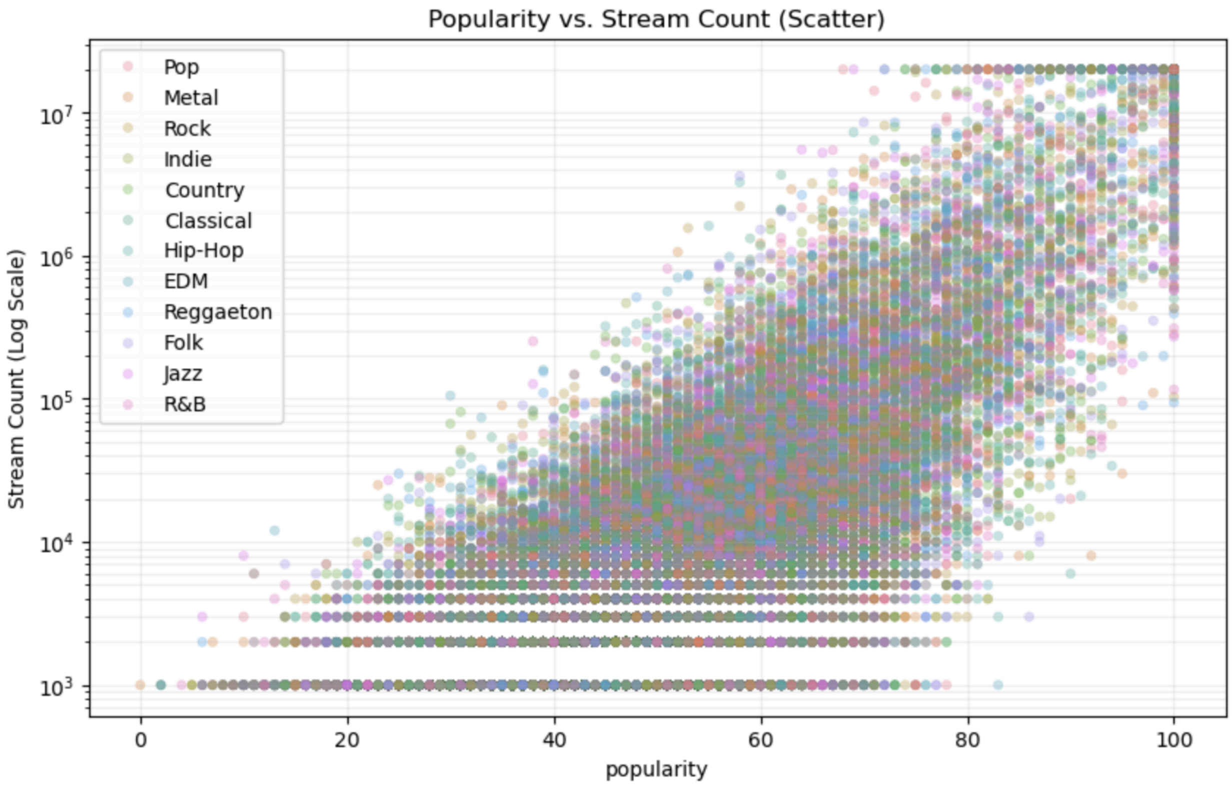
Label - As a target client for this model, understanding how labels are represented and how they impact this data is vital to this project. In total, there are 8 labels that represent all of the songs in this dataset. When exploring how they ranked between total stream count, number of artists represented, and number of songs represented, and mean_popularity there was not one label that ranked 1st in more than one category indicating a good variety in labels within out dataset.

Stream Count - While exploring Artist, we learned that the stream count column maximum value of streams of 20 Million, and that value was applied to 458 records indicating that it is a limit. We can assume that songs with a value of 20 Million Stream Count likely had more streams, and, since the songs in this dataset have release dates spanning 10 years, some songs have had more time to reach their number of streams. This indicates that stream count is not a reliable source to determine popularity, plus with the objective of this project being to determine song popularity prior to release, training the model on stream count could create a reliability on data that will not be present in practical use.

Songs - While exploring the dataset for metrics on songs, I identified that there were 201 songs with both a perfect popularity score of 100, and a maximum streaming count of 20 Million. While reviewing visualizations of the popularity and stream counts for songs, it is also clear that the vast majority of songs have less than 2.5 million streams. Comparing popularity to stream count does show that there is a positive correlation between the two, which was expected but not as obvious given the somewhat surprising results with the mean popular ratings vs number of streams. Noting that there was an advantage to stream count for songs that were released earlier, I dug deeper into the release date and stream count. A



scatter plot showed a relatively uniform distribution between time since release date and total streams, however inspecting the average streams per day showed that there were many songs that had a significantly higher streaming per day ratio.



Genre - Exploring Genre followed many of the same trends as the other features. The total number of streams and popularity scores did not fully match. The analysis indicated that the pop genre was the most popular but was the 6th most streamed genre, whereas hiphop was the most streamed but the 4th most popular. At first glance this could be misleading, indicating that there is some sort of skew in distribution, however a scatter plot visualization (above) utilizing the hue by genre indicates that while there is a linear positive correlation between popularity and stream count, the distribution of genre is homogeneous.

Region/Country - Exploring the popularity and stream counts by region and country indicated that Asia is the top region for both popularity and stream count, making it an ideal target for song writers.

After completing manual exploration of these features, I created a **Pearson Correlation Heatmap** of the numerical features of the dataset. It showed that no single numeric feature had a strong linear relationship with track popularity with the exception of stream count which is the other target feature of the project, suggesting that the track popularity is not driven by simple linear effects.

I then performed **Categorical Feature Correlation Analysis**. This analysis also indicated that there was no singular relationship with track popularity, again indicating that there is not a linear or singular driver to song popularity.

In conclusion of the EDA step, all features will be kept while entering the pre-processing step to ensure the model has sufficient data to utilize in training.

Pre-Processing

Standard steps for Pre-Processing were followed in this step. After the dataset was loaded, several columns were identified that needed to be dropped due to redundancy, high cardinality, or leakage risk. Columns dropped were:

- **track_id** - This is a unique ID used to identify the track in the dataframe. It is not the name of the song that will be released and will not be public knowledge.
- **track_name** - This is the actual name of each song and has a very high number of unique values.
- **artist_name** - In EDA we discovered that the dataset is not heavily influenced by artist name for songs, the majority of artists only have one song and therefore one record. Due to potential high-cardinality issues, this column will be dropped.
- **album_name** - In the EDA phase we found that most albums only contain one song so there are a very high number of unique values.
- **duration_mm**:ss:ms - This column contains the same information as duration_ms only in a different format that is easier to read. It was created only to improve the EDA process.
- **latlng** - This column was kept for EDA, it contains a list version of the latitue and longitude of each country. It is no longer needed since we have the country name.
- **capital** - since our dataset does not contain insights of number of streams or popularity per city, Capital is essentially a duplicate feature of Country
- **demonym** - This feature is redundant with Country
- **stream_count** - Stream Count poses risk to leakage and training the model on a feature that will likely not be present during practical use.
- **Release_date** - Release date was converted to a column 'days_since_released' which was calculated from the release date column. Days since released will be a better to train the model, and renders release_date unnecessary

Once these columns were dropped from the dataset, I performed One Hot Encoding on categorical columns using pandas get_dummies dropping the first to avoid multicollinearity. Then the dataset was split into 'X' and 'y' variables with the target 'popularity' assigned to 'y' and dropped from 'X'. Once split, I utilized sklearn's train test split to split the variables into training and test sets with an 80/20 ratio. Feature scaling was applied after the train-test split to prevent data leakage. The scaler was fit only on the training data and then applied to both the training and test datasets to ensure that the model evaluation reflects real-world performance.

Modeling

Three models were trained and tested in the modeling step. Each was selected for specific reasons and diversity in how the models process the data. This allowed me to evaluate if increasing modeling complexity improved predictive power.

Linear Regression - Selected as a foundation model to detect if there are strong linear relationships present.

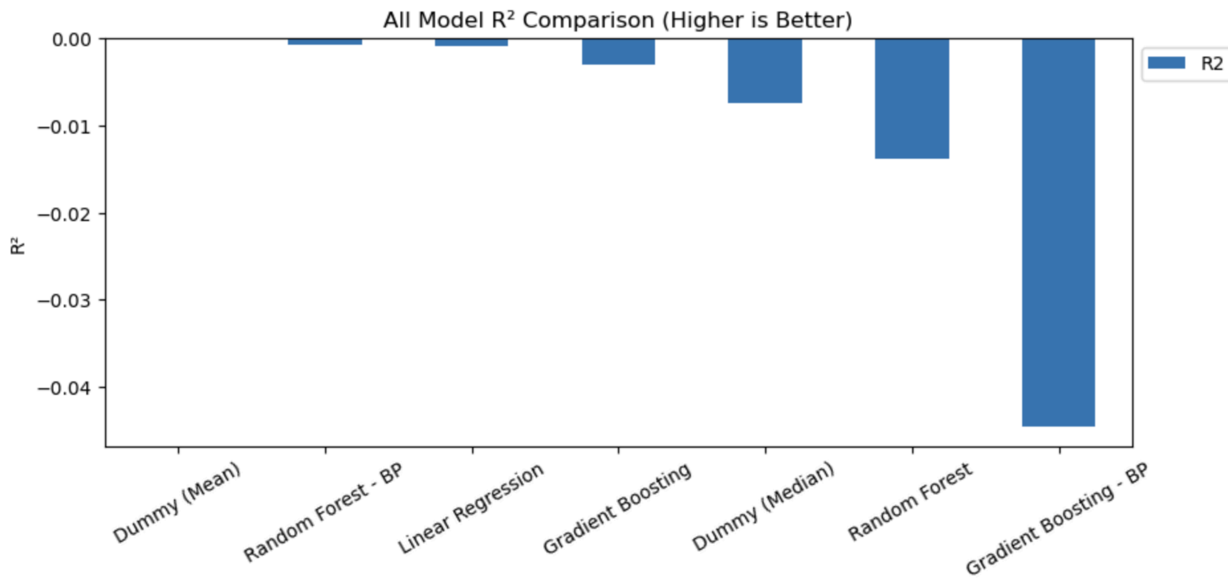
Random Forest Regression - Selected to capture non-linear relationships and interactions

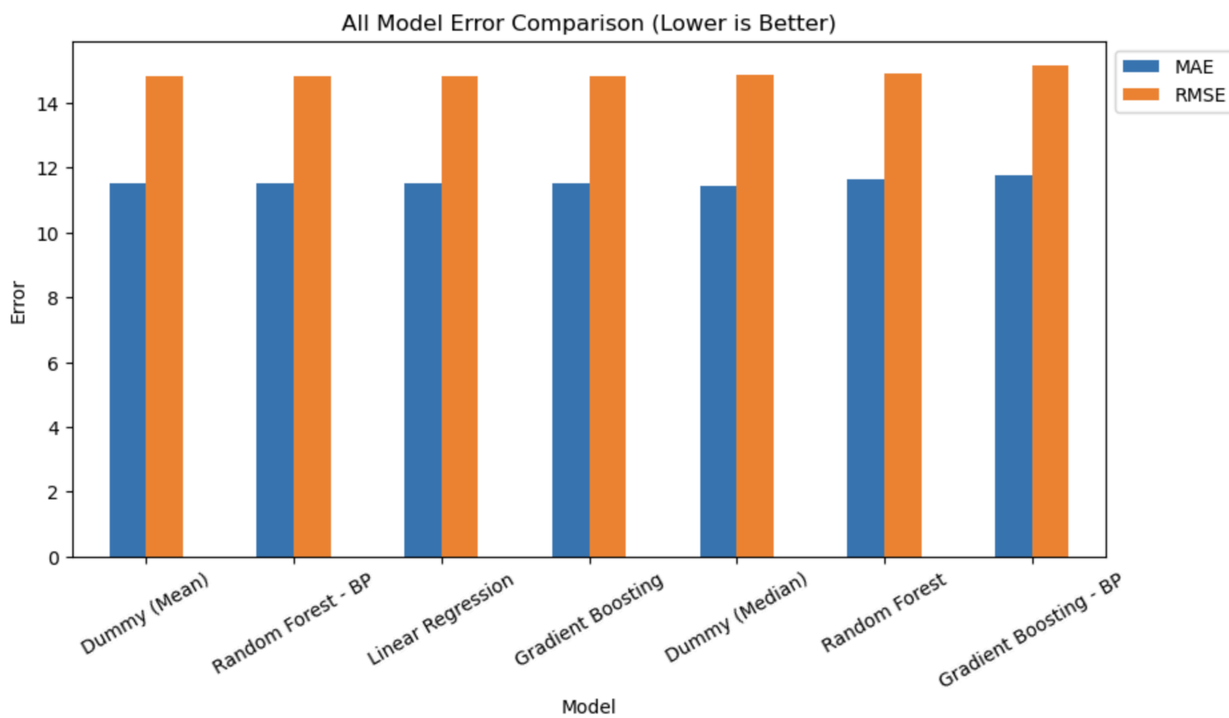
Gradient Boosting Regression - Selected to test a high-capacity ensemble model

To establish a baseline, two DummyRegressor models (mean and median strategies) were added. These models predict the average popularity for all songs without learning from features.

After training each of the models, they were tested and compared to evaluate the results. The initial models showed negative scores on the R Squared metric. Hyper Parameter tuning was performed using GridSearchCV and the best parameters identified were applied to Random Forest and Gradient Boosting models, while this improved the results, the R Squared remained negative. In fact, the highest performing model was the DummyRegressor Mean model, indicating that the models failed to explain variance in popularity beyond simply predicting the average. **This strongly suggests that the available pre-release audio and descriptive features do not contain sufficient predictive signal to model song success.**

MODEL	R2	MAE	RMSE
Dummy (Mean)	-0.000097	11.512126	14.826212
Random Forest - BP	-0.000753	11.519009	14.831073
Linear Regression	-0.000924	11.515795	14.832340
Gradient Boosting	-0.003130	11.530684	14.848673
Dummy (Median)	-0.007459	11.455653	14.880678
Random Forest	-0.013791	11.639644	14.927369
Gradient Boosting - BP	-0.044528	11.788452	15.151972





Alternative Approaches

Several alternative modeling strategies were explored to ensure a comprehensive evaluation of the dataset. First, the target variable was converted into a binary “hit” classification problem using various popularity thresholds. However, due to class imbalance, classification models defaulted to predicting the majority class, yielding misleading accuracy but near-zero precision and recall. Attempts to engineer additional artist-level features introduced leakage or failed to improve performance. Finally, a pivot was made to predict the country in which a song would perform best, but removing country-level data to avoid leakage left insufficient signal, and model performance remained poor.

Final Conclusion

Extensive modeling, hyperparameter tuning, baseline comparison, and alternative strategy exploration have led us to a clear conclusion: this dataset does not contain enough correlating pre-release features to accurately predict song popularity.

The fact that the trained models did not outperform a simple DummyRegressor baseline provides strong evidence that song success is driven by external factors not captured in this dataset. Access to additional features such as artist popularity, trends in genre popularity by country or region, language of the song, and user trends by country/region could help improve performance. Additionally, this dataset contained several features that did not provide enough information to narrow down to specific insights. The length of time captured over the course of 10 years led us to have to make assumptions about stream count. The high number of artists with only one song included in the dataset did not allow us to truly observe the relationship between artist popularity and song popularity. Only providing the single country that the song was successful in did not allow us to truly understand if song popularity is generally global or specific to Country. And most importantly, popularity on a scale of 0-100 did not provide insight on how popularity score was determined in the first place, especially when considering there are many songs in the dataset with a perfect 100 popularity score, but less than 20 Million streams. Expansion to these features could have a significant improvement the models performance.

The outcome of this project is not a deployed model, but a clear, data-supported determination of the dataset's limitations — and a roadmap for what additional signals would be required to meaningfully predict musical success.